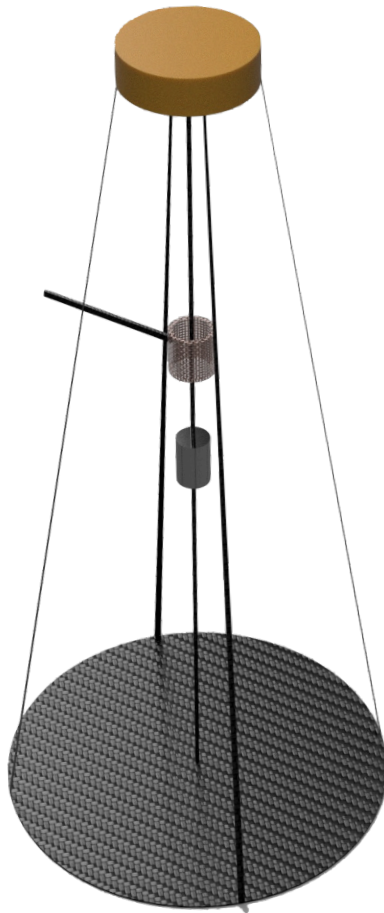


MAE / ECE 535: DESIGN OF ELECTROMECHANICAL SYSTEMS

Design Project

Jack Fowler, Vincent Wu

April 28, 2026



Abstract

This report details the electromechanical design of a portable, 1 kg direct-drive Wave Energy Converter to provide continuous smartphone charging (5V at 500mA). To achieve resonance with target wave periods, a heave plate was implemented to leverage fluid mass. System kinematics were analytically modeled in MATLAB, and the magnet coil system was optimized via ANSYS Maxwell. The finalized device, comprising a 40x60 mm N52 magnet and a 990-turn stationary coil, demonstrated a peak output of 50.12 W and generated 3.764 J per stroke. Circuit simulations verified that a rectified capacitor buffer successfully smoothed these energy pulses into a steady 2.5 W. While the design successfully meets theoretical electrical generation requirements, future work is necessary to address idealized magnetic assumptions and validate the structural integrity of the heave plate assembly under drag forces.

Introduction

There exists an opportunity to harness waves for energy generation. This document will detail the design of a portable, permanent magnet based electromechanical Wave Energy Converter (WEC) for the purpose of charging smartphones. To meet usage and portability requirements, the device will have the following constraints: 1kg total weight and produce a 5V output at 500mA^[2]. Because the total mass is so small, the device cannot rely on absorbing the energy from large, robust waves. To ensure survivability and feasible power generation, the following operational criteria are established:

- Wave Height Restriction: The device must operate in moderate wave heights between 0 m and 2.0 m, avoiding extreme offshore conditions that would destroy a small-scale, portable structure.
- Frequency Range: To ensure sufficient energy buffering between wave crests, the device targets highly active wave periods ranging from 1.0 s to 5.0 s.

After research, coastal piers, jetties, and similar abutments fit best to this operational profile. They provide accessible structural attachment points, experience moderate near-shore weather conditions, and are locations where grid power is typically inaccessible for small charging tasks, making the WEC highly practical for the end-user.

To simplify the problem and allow for reasonable analytical and computational modeling, the following assumptions will be made:

- 1) Harmonic forcing: Ocean waves are irregular, but can be well approximated by monochromatic waves, whose average periods and amplitudes are available.
- 2) Linear wave kinematics: For our device's target conditions, the wave amplitude will be small compared to the wavelength.
- 3) Pure heave motion: This WEC will be modeled as a single degree-of-freedom oscillator, assuming our attachment method only allows purely vertical motion.

To conduct a case study, historical environmental data were extracted from the National Laboratory of the Rockies Marine Energy Atlas^[3]. The Nags Head Fishing Pier in the North Carolina Outer Banks was selected as an ideal representative deployment site. The average near-shore wave characteristics for this location are:

Wave Power: 1.610 kW/m
Wave Height: 1.002 m
Energy Period: 3.02 s
Spectral Width: 0.523



Figure 1) Marine Energy Atlas Interactive Data Tool

Methods

Decision Matrix

Device design began with research into various architectures that can convert wave energy to electrical power. Two types were considered: direct-drive linear generators and rotary generators with mechanical translators.

Table 1) Design Matrix for Device Selection

Generator Type	Description	Pros	Cons
Direct drive linear	Converts the harmonic wave motion via a magnetic rod moving linearly through a coil of wire.	Simple, fewest moving parts No mechanical transmission losses High reliability	Requires a large PM Low induced voltage
Rotary rack and pinion	Converts the wave motion into rotational motion. A rack moves up and down, spinning a circular gear that drives a rotary generator.	Higher magnetic field velocity, better voltage. Existing DC generator use. Possible flywheel	Complex Frictional losses High mechanical wear Heavy components
Rotary “firestarter” generator	The rotary generator sits in the oscillating buoy, with cords attached to the top and bottom, translating the harmonic motion through pulleys, similar to a stick and string firestarter.	Less weight than other radial rectifiers, similar benefits.	Complex Risk of tangling
Horizontal waterwheel	A floating cylinder fitted with paddles or blades that the waves pass by and push, spinning the wheel to drive a generator.	Simple design	Difficult to model horizontal shore waves Bulky inefficient geometry
Water column-driven turbine	A hollow chamber is open to the wave motion, and through rise and fall, air is pushed in and out, spinning a turbine.	No generating parts in contact with seawater.	Complex Requires a large air chamber

The options were considered, and quickly the difficulty of implementing the complex mechanism heavy, rotary, waterwheel, or turbine generator was deemed too extreme for our allowable mass budget and operating conditions. The pure heave buoy “firestarter” generator was closely considered. For power generation, it would have superior power output versus a direct linear one, as the simulation in [Appendix G](#) shows. However, the complexity of mechanisms and the fragility of components remove it as a viable candidate. The unpredictable, outdoor, corrosion-heavy operating environment, paired with the mass budget, made the direct drive linear generator the best choice to move forward.

Motion of a Pure Heave Buoy

Since our device relies on the oscillatory motion produced by ocean waves, the physical relationships between the ocean waves and the electromechanical systems were first established. The electrical power produced by a direct-drive linear generator is proportional to the square of the velocity at which the magnetic field is traveling through the coil. From

Faraday's Law and Lorentz forcing, the electrical power generated is $P = cz^2$, where c is the linear damping coefficient

and $\dot{z}$ is the vertical heave velocity of the buoy^[1]. Thus, to optimize power, our harmonically oscillating buoy should achieve maximum velocity. This occurs when the frequency of the driving wave matches the natural undamped frequency of the oscillator, known as resonance.

Based on our 3.02s average wave period (T_e), the forcing frequency of our wave can be found as $\omega = \frac{2\pi}{T_e} = 2.08 \text{ rad/s}$.

The natural frequency of a floating body in heave is defined by the ratio of its restoring force to its inertial mass^[1]:

$\omega = \sqrt{\frac{k_{hs}}{m}}$, with the hydrostatic stiffness $k_{hs} = \rho_{seawater} g(A_w)$, the product of seawater density, gravity acceleration, and the cross-sectional area at the surface.

For the buoy case, this stiffness relationship introduces a paradox when attempting to frequency match and induce resonance. To float the 1 kg mass, a cylindrical buoy was selected with an outer diameter larger than 100mm. This inherently produces a natural frequency far too high to resonate. To resolve this, a [heave plate](#)^[4], a thin, flat, rigid disk suspended deep beneath the wave surface, can be used. As this disk accelerates, it moves the surrounding fluid, adding some of the water around it to our effective mass, increasing the denominator term of our natural frequency. The added mass for a thin cylinder, calculated from potential flow theory, is $m_{heave} = \frac{8}{3} \rho_w R_{heaveplate}^3$.

The dynamic response of our WEC was modeled as a single-degree-of-freedom oscillator in heave. The foundation of this analytical model was found by moving through Newton's Second Law of Motion

$$(m_{total}) \ddot{z} = F_{excitation}(t) + F_{hydro}(z) + F_{generator}(z, \dot{z}) + F_{drag}(z, \dot{z}, t)$$

- The wave excitation force was modeled using the Froude-Krylov approximation, neglecting diffraction effects due to how small our object is^[1]. This approximation describes the force acting on the bottom of the buoy, driven by a dynamic pressure field from the wave, and is expressed as the product of hydrostatic stiffness and instantaneous wave surface elevation $\eta(t)$, which is simply harmonic for our wave. $F_{excitation} = k_{hs} A \cos(\omega t)$
- The hydrostatic buoyant force^[1] for our pure heave buoy is $F_{hydro} = -(\rho_{seawater} g A_w)$
- When our linear generator is engaged, a mechanical damping force acts against the buoy $F_{generator} = -c \dot{z}$. Our damping coefficient for this system is not constant, since our magnet is moving in and out of our coil, and is modeled $c = \frac{(B_{avg} L_{act})^2}{R_{total}}$, with B_{avg} being an average electric field intensity in our coil, L_{act} being the length of the magnet engaged by the coil, and R_{total} being our total circuit resistance. This is only applied when the magnet is passing through the coil.
- Viscous damping was modeled using the Quadratic Morison Equation^[6], dictating drag by the relative velocity of the moving buoy and the surrounding water particles $F_{drag} = -\frac{1}{2} \rho_w C_d A_p (\dot{z} - w)|\dot{z} - w|$, with C_d being the drag coefficient for the respective object, A_p being the projected area, and w the moving water. Since water particle velocity varies in moving waves, Airy Linear Wave Theory^[6] is used to determine the velocity at each relevant depth for the buoy and heave plate through motion. $w(z, t) = A \omega^{kz} \sin(\omega t)$.

This model was time-stepped through and evaluated using a Runge-Kutta numerical integration in MATLAB, producing our magnet motion, which was then used to calculate approximate power output. The motion and output graphs can be seen in [Appendix A](#).

Power Generation

The nature of this linear generator lends itself to producing power in bursts, with periods of little to no power in between. With this in mind, the energy needed by the load (E_{load}) per time period (T) and the energy generated (E_{gen}) per burst lasting (t_{on}) can be described by the following equations.

$$E_{load} = P_{load} \cdot T \quad E_{gen} = P_{gen} \cdot t_{on}$$

Rearranging and solving for the needed generated power in watts (P_{gen}), it is shown that the generated power is proportional to the power load (P_{load}) and the duty cycle ($\frac{T}{t_{on}}$).

$$P_{load} \cdot T = P_{gen} \cdot t_{on} \quad P_{gen} = P_{load} \cdot \frac{T}{t_{on}}$$

Since the load and wave characteristics are known, the unknown variables to optimize are the power that needs to be generated and the duration that the power must last. First, reviewing the governing equations for producing power using electromagnetism, Faraday's Law states that the voltage induced on a coil (V_{ind}) is proportional to the number of turns in the coil (N) and the rate of change of magnetic flux (Φ_b) through the coil. Magnetic flux can be found by the magnetic flux density (B) normal to an area (A).

$$\Phi_b = B \cdot A \cdot \cos(\theta) \quad V_{ind} = -N \cdot \frac{d\Phi_b}{dt}$$

Since the air gap between the coil and the magnet should be kept to a minimum to reduce magnetic reluctance, and the velocity of the magnet is determined by the ocean waves, the two most important parameters left are the magnetic flux density and the number of turns in the coil.

To optimize these parameters, a MATLAB program was developed to find a rough range of magnet sizes and coil sizes that will produce the necessary power. A N52 neodymium was chosen due to its high magnetic flux density of 1450mT ^[7], and 24 AWG magnet wire was chosen for its 3.5A carrying capacity^[8] and high packing density. The program uses the previously calculated motion and approximates the induced voltage, current, and power generated as the magnets move through the coil. ([Appendix A](#)) The program was swept through a range of magnet sizes and coil sizes to find a functional pairing. Since the mass of the device is limited to 1kg, 800g was initially budgeted towards the coil and magnets, with 200g being reserved for structural items, the air-filled buoy, and the heave plate.

Once the range of magnet and coil sizes was narrowed down, ANSYS Maxwell was used to further refine the parameters and improve the accuracy of the power generation.

ANSYS Maxwell Simulation

First, various coil heights were swept to find the duration that power is produced and the magnitude of power. This showed that while both the time on and the power generated increased with the height of the coil, neither changed drastically. The duration that the magnet is close enough to produce power only differed by 0.06 seconds, and the power produced differed by 10.6W between the tallest and shortest coils. Knowing that the duration the magnet is within range is from 0.18 seconds to 0.24 seconds and that the period of the waves is 3.02 seconds, the range of power needed can be estimated using the formula derived earlier. This results in a minimum power range of 31.46W to 41.94W. Since the

power is only produced for a fraction of the time the magnet is in range, this estimation will be on the low end of the required power. ([Appendix C](#))

When the number of coil layers was swept, the amount of power changed greatly. A difference of more than 40W was seen between having 8 coil layers and 13 coil layers. With this information and the limit of 1000 turns total, a coil height of 90 turns and 11 layers was chosen to maximize both the duration of the power pulses and the amount of power. In total, the coil has 990 turns with an internal resistance of 12Ω . This also results in a pulse duration of 0.19 seconds, and therefore a minimum power requirement of 39.74W. ([Appendix C](#))

With the size of the coil decided, the height of the magnet was swept next. The magnet height also has a massive impact on the power, with a difference of 44.49W between a 10mm tall magnet and a 60mm tall magnet. Since it is within the weight budget, and a minimum power generation of 39.74W is required, the 60mm tall magnet was chosen.

A final simulation was run using the optimized parameters. As seen in [Appendix B](#), the peak power is calculated. To refine the power estimations, the required energy to charge the smartphone is calculated. With a power draw of 2.5W and a wave period of 3.02, 7.55J is needed every wave to sustain the power draw. Since the magnet passes through the coil twice every period, the energy required from the generator is 3.775J. In ANSYS, using the cumulative integral function on the power produced by the coil, the total energy produced by the coil can be graphed ([Appendix B](#)).

Power Rectification

The power produced by the linear generator is not constant and flips polarity as the magnet passes through the coil. For this generator to charge a device, it must output constant DC power. To solve this problem, a circuit must be built that can rectify the power, store power in between pulses, and maintain a steady voltage.

Firstly, rectification of the power can be achieved via a full bridge rectifier^[9]. This consists of four Schottky diodes arranged such that current can only flow in one direction. Thus, allowing the positive half of the power to pass through, and flipping the negative half.

To maintain power while the generator is not producing electricity, capacitors are placed at the output of the full bridge rectifier. The capacitors will store the excess energy produced by the generator, then discharge the energy between power cycles. This both maintains power throughout the wave period and also smoothes out the pulses in power.

Finally, a boost-buck converter can be used to hold a steady 5V output. Buck-boost converters can shift any voltage within its operating range to the desired voltage. For this generator, the TPS63070 by Texas Instruments^[10] was chosen. This converter can maintain 5V from a 2V to 16V input with a typical switching resistance of $50m\Omega$ to $100m\Omega$.

To validate this circuit design, Falstad^[11], a circuit simulation tool, was used ([Appendix E](#)). To simulate the power from the generator, two voltage sources were placed with their polarities opposite each other. These were set to have pulses at 0.66Hz (doubled frequency of wave since there are two pulses during a period of the wave), a duty cycle of 15.27% (calculated from [Appendix B](#)), and a max voltage of 24.5V. A 12Ω resistor is placed in series with the sources to simulate the resistance of the coil. The sources are then fed into the full bridge rectifier made using Schottky diodes. A 1F capacitor is placed on the output of the rectifier. A 10.1Ω resistor is used to simulate the load on the generator. 10Ω is derived from Ohm's Law and the known charging voltage and current supplied. 0.1Ω is added to simulate the switching resistance of the buck-boost converter.

Results

The analysis and simulation with MATLAB, ANSYS Maxwell, and Falstad successfully demonstrate the feasibility of a 1kg linear WEC, theoretically capable of the minimum charging for a smartphone.

To achieve resonance with the 3.02 s wave period, the mechanical system was successfully iterated to parameters that fit within the mass budget: a 40mm x 60mm magnet mass estimated at 0.562 kg, and the 990 turn coil at 0.317 kg. The remaining 0.127kg was distributed across the structural and electrical components. A 486mm heave plate was successfully added to the bottom of our oscillating 150mm buoy, increasing our effective mass by 39.22kg, allowing us to reach resonance with the wave, maximizing the velocity up to a peak of 0.84 m/s. An approximate 50V at 40W was predicted analytically by integrating the instantaneous power as the magnet passes the coil. ([Appendix A](#))

ANSYS Maxwell was successfully utilized to more accurately simulate the magnet harmonically moving through the coil, producing a peak voltage of 24.52V at 2.04A (50.12W) with a 0.19s pulse duration ([Appendix B](#)). Integration of the ANSYS power curve yielded 3.764 J per pass of the magnet, nearly the minimum target energy requirements defined in the methods.

Finally, a circuit simulation was run ([Appendix E](#)), verifying that the 1 F capacitors, paired with a buck-boost converter, effectively buffered the pulses out for charging. In roughly 1 minute, the circuit was charged to 5V and reached steady state. Over a 15 minute simulated run time, the circuit supplied uninterrupted 5V at 500mA to a simulated 10 ohm charging smartphone.

The mass budget of 1kg was met and was validated by CAD approximations ([Appendix E](#))

Conclusions

The design and evaluation of the portable WEC was theoretically successful, demonstrating that a 1.0 kg direct-drive linear generator can feasibly meet the 2.5 W (5V at 500mA) continuous charging constraint in a near-shore environment. This project leveraged relevant course topics, including energy conservation, Maxwell's equations, magnetic circuit analysis, numerical integration, FEM, and more.

While the necessary energy was achieved in simulation, it is likely that these idealized models overpredict real performance. The initial analysis script made several assumptions, such as averaging the magnetic flux, neglecting fringing and reluctance, and assuming a rigid device. The ANSYS Maxwell setup used an ideal motion track, neglecting the deceleration in the coil. The Falstad circuit simulation doesn't account for efficiency losses as the circuit heats up during charging. All of these simplifications will produce real-world inaccuracies.

The physical device is also likely to structurally fail under real use. While certain high-performance materials, like CFRP, were approximated as structural members in the mass budget, significantly more analysis and testing would be required to ensure the stability of the device under operation.

This project suggests the mathematical and electromagnetic viability of a portable, direct-drive WEC, but it is far from rigorously evaluating the reality of constructing one.

References

- [1] J. Falnes, *Ocean Waves and Oscillating Systems: Linear Interactions Including Wave-Energy Extraction*. Cambridge, U.K.: Cambridge University Press, 2002.
- [2] USB Implementers Forum, *Universal Serial Bus Specification Revision 2.0*. USB-IF, 2000.
- [3] National Laboratory of the Rockies, "Marine Energy Atlas Data Viewer," *Marine Energy Atlas*. [Online]. Available: <https://maps.nlr.gov/marine-energy-atlas/data-viewer>.
- [4] ScienceDirect, "Heave plate," *ScienceDirect Topics*. [Online]. Available: <https://www.sciencedirect.com/topics/engineering/heave-plate>
- [5] S. Chakrabarti, *Handbook of Offshore Engineering*. Amsterdam, Netherlands: Elsevier, 2005.
- [6] R. G. Dean and R. A. Dalrymple, *Water Wave Mechanics for Engineers and Scientists*. Singapore: World Scientific, 1991.
- [7] Arnold Magnetic Technologies, "Sintered Neodymium-Iron-Boron Magnets," Neodymium N52 Datasheet. [Online], Available: <https://www.arnoldmagnetics.com/wp-content/uploads/2017/11/N52-151021.pdf>
- [8] Cambion, "Single Pole Sockets Performance Data." [Online]. Available: http://www.cambion.com/single_pole_sockets_performance_data.aspx
- [9] AspenCore, "Full wave rectifier and bridge rectifier theory," *Basic Electronics Tutorials*, Aug. 9, 2024. [Online]. Available: https://www.electronics-tutorials.ws/diode/diode_6.html
- [10] Texas Instruments, "TPS63070 2-V to 16-V Buck-Boost Converter With 3.6-A Switch Current," TPS63070 Datasheet, Jun. 2016 (Revised Mar. 2019). [Online]. Available: https://www.ti.com/lit/ds/symlink/tps63070.pdf?ts=1777430881864&ref_url=https%253A%252F%252Fwww.google.com%252F
- [11] Falstad, "Circuit Simulator Applet." [Online]. Available: <https://www.falstad.com/circuit/circuitjs.html>

Appendices

Appendix A) MATLAB Outputs

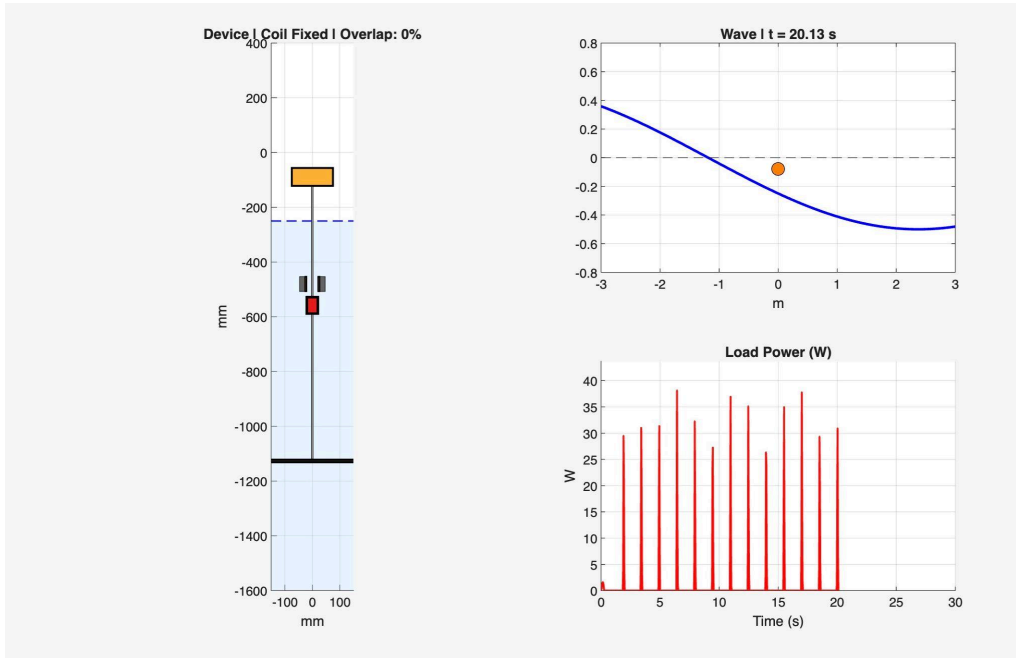


Figure A1: Screenshot from animated output containing moving visual, wave bobbing, and output power approximation.

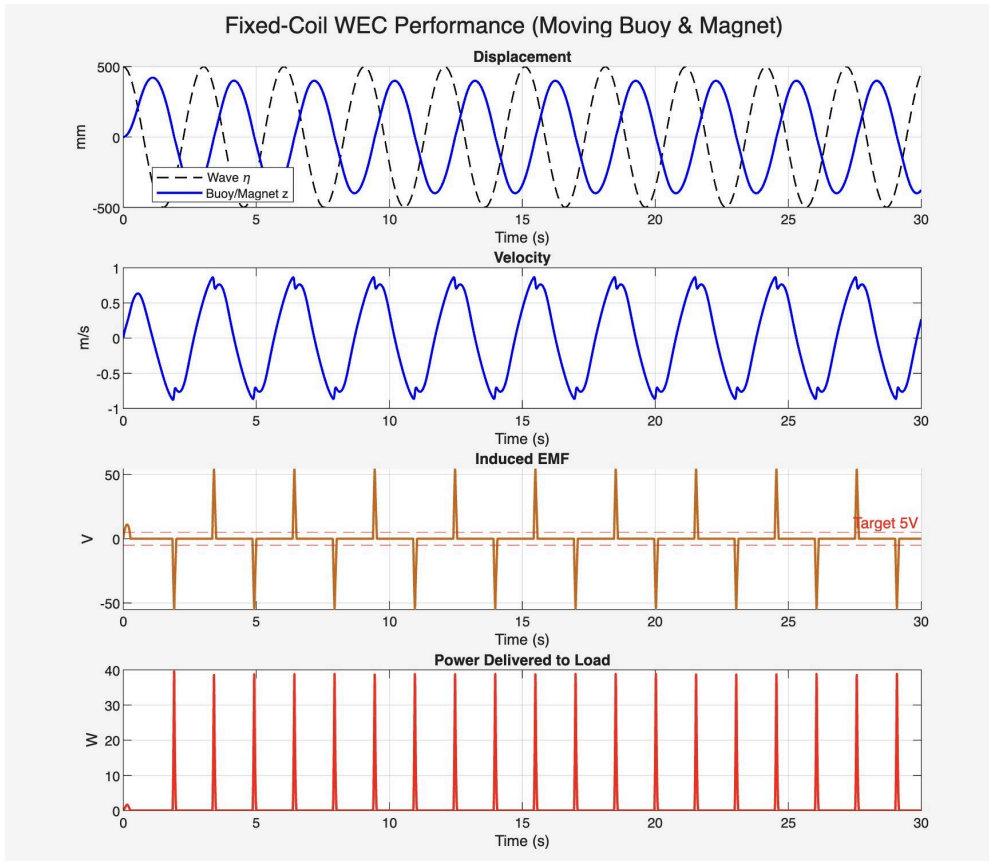


Figure A2: Output graphs of buoy displacement, velocity, induced voltage, and output power.

Appendix B) Coil Power Production in ANSYS Maxwell

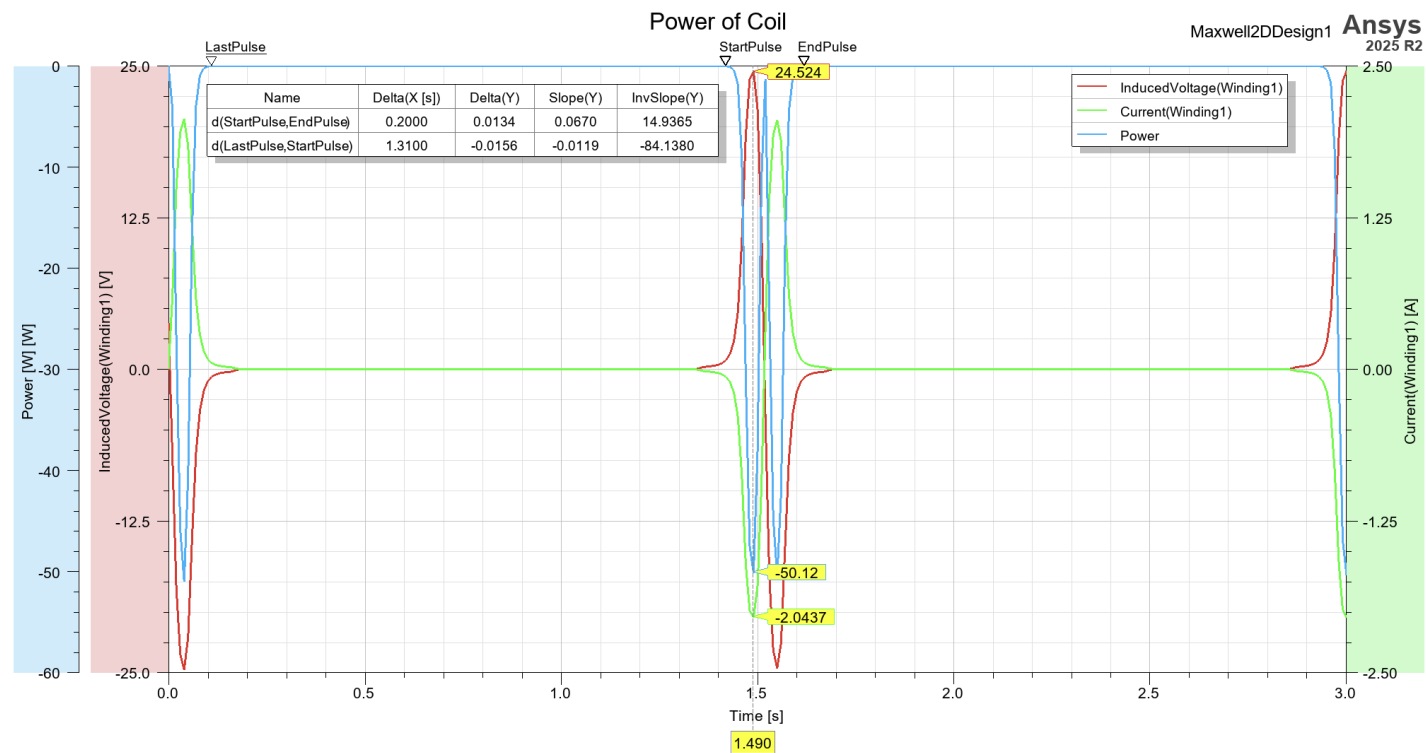


Figure B1: Graph displaying the peak power, voltage, and current produced by the optimized coil and magnet setup. Includes time deltas for the duration of power generation pulses and the period between power pulses.

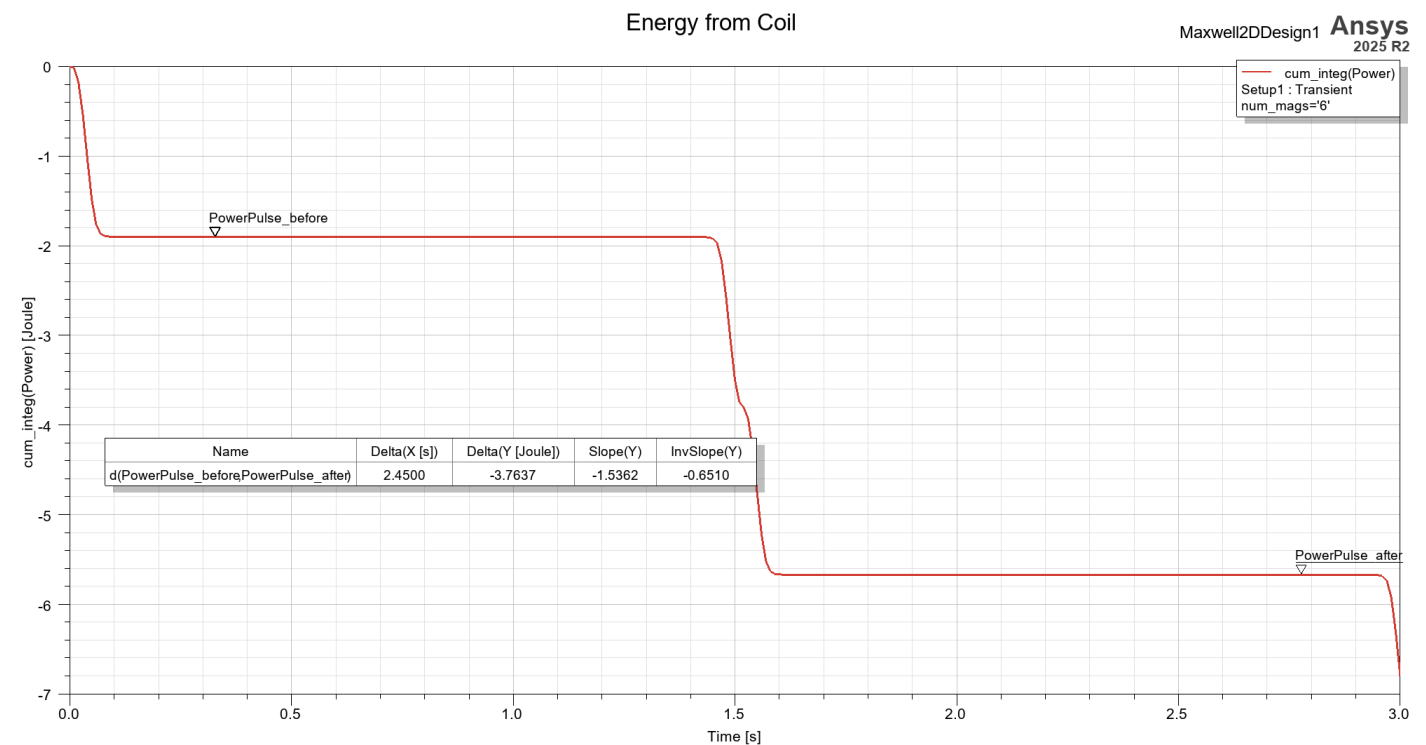


Figure B2: Graph displaying the accumulated energy in Joules of the direct-drive linear generator. The table shows the Joules generated from one pulse.

Appendix C) Parametric Sweeps in ANSYS Maxwell

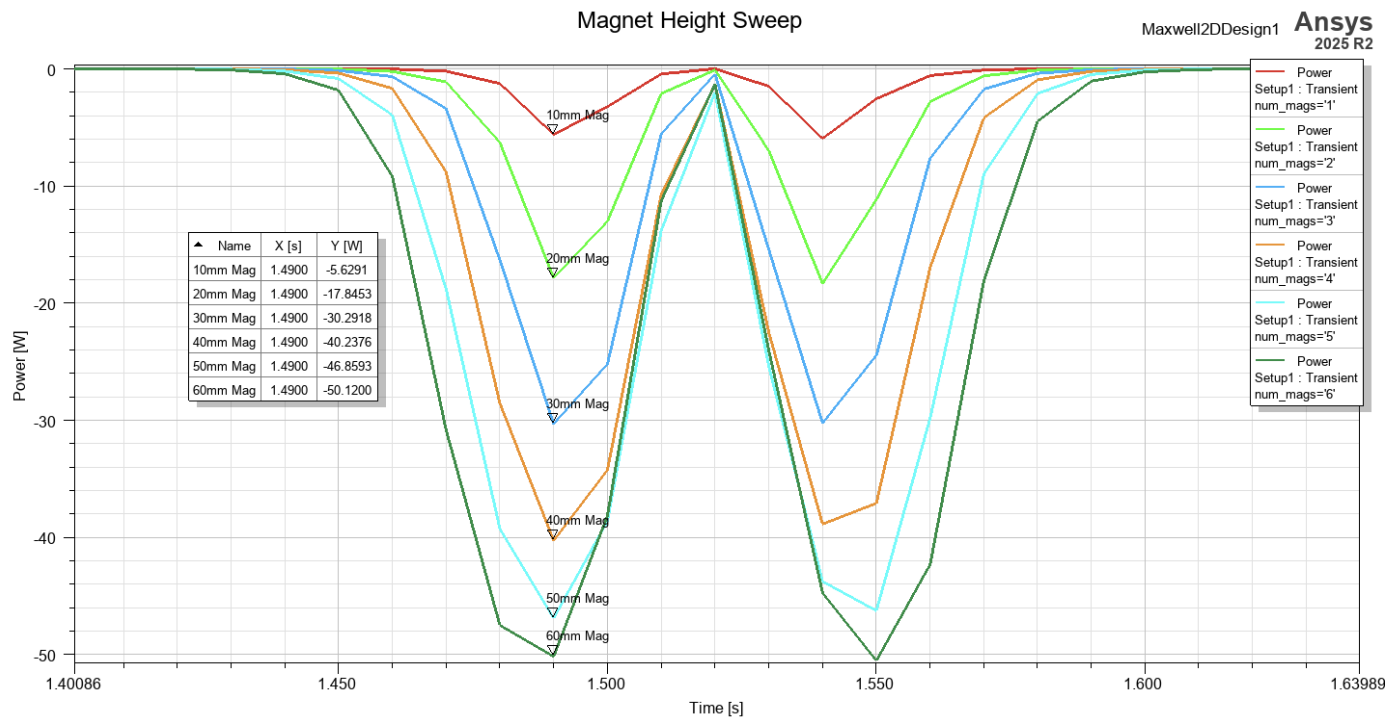


Figure C1: Graph displaying power generated when magnets of 6 different heights pass through the coil. The table shows the peak wattage from each magnet.

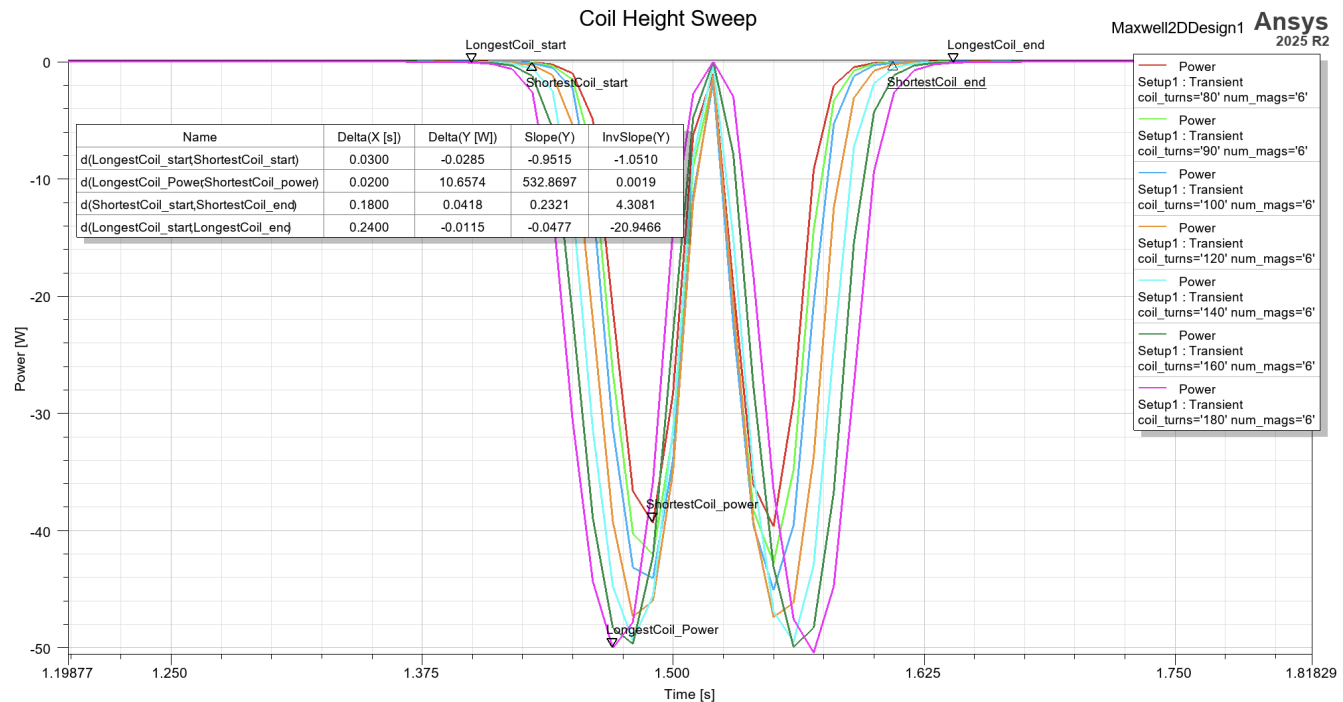


Figure C2: Graph displaying the power generated when a magnet passes through 7 different coil heights. The table shows the time delta between when the longest coil starts generating power and when the shortest coil starts generating power, the duration of the longest power generation, the duration of the shortest power generation, the peak power of the longest coil, and the peak power of the shortest coil.

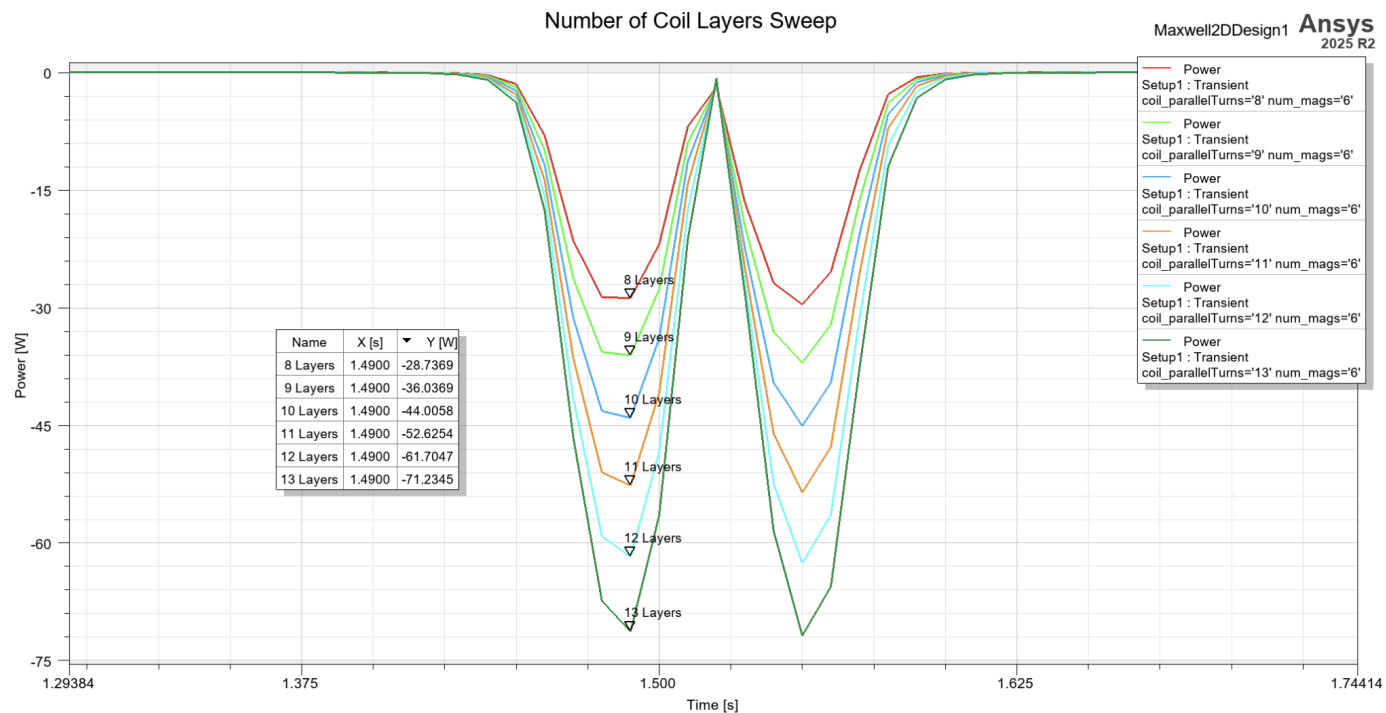


Figure C3: Graph displaying the power generated by coils containing 6 different numbers of layers. The table shows the peak power for each coil.

Appendix D) Magnet Field Plot in ANSYS Maxwell

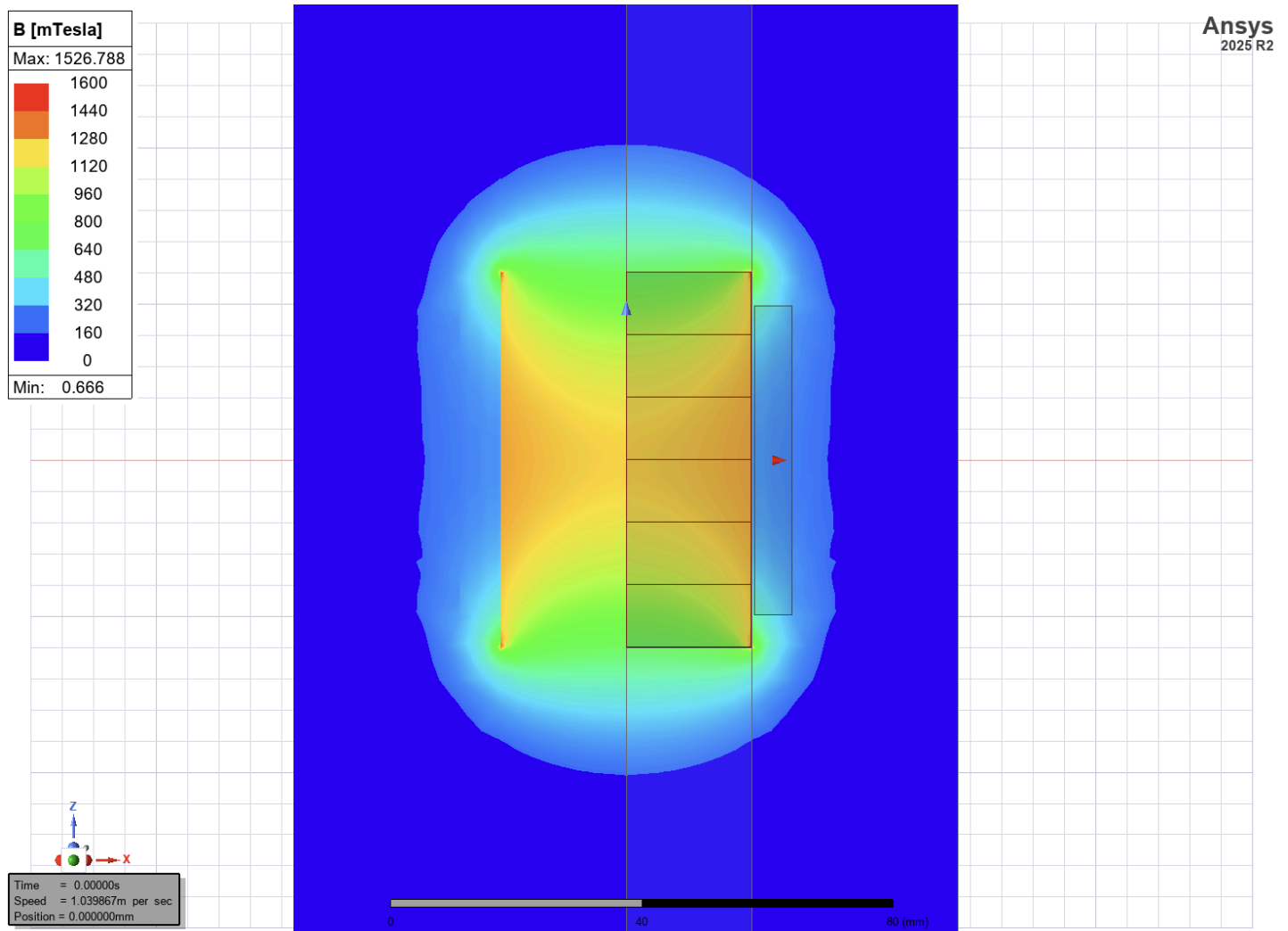


Figure D1: Magnetic field visualization in ANSYS Maxwell, demonstrating field intensity at the coil.

Appendix E) Rectifying Circuit Simulation

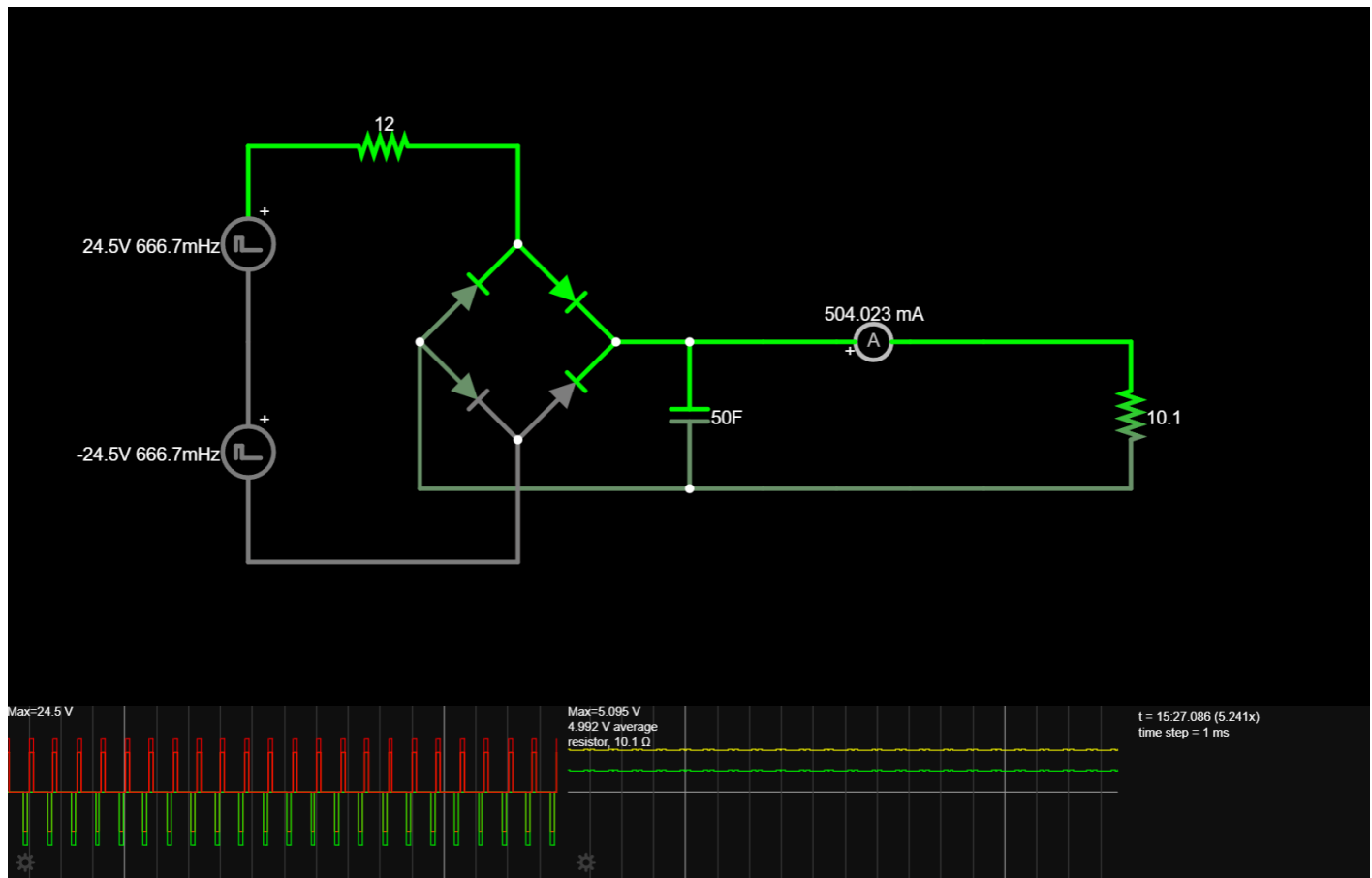


Figure E1: Falstad circuit simulation of a filtered full bridge rectifier, driven by an approximation of the power produced by the direct-drive linear generator. A resistive load representing charging a smartphone is placed at the outputs.

Appendix F) CAD Mass Measurement

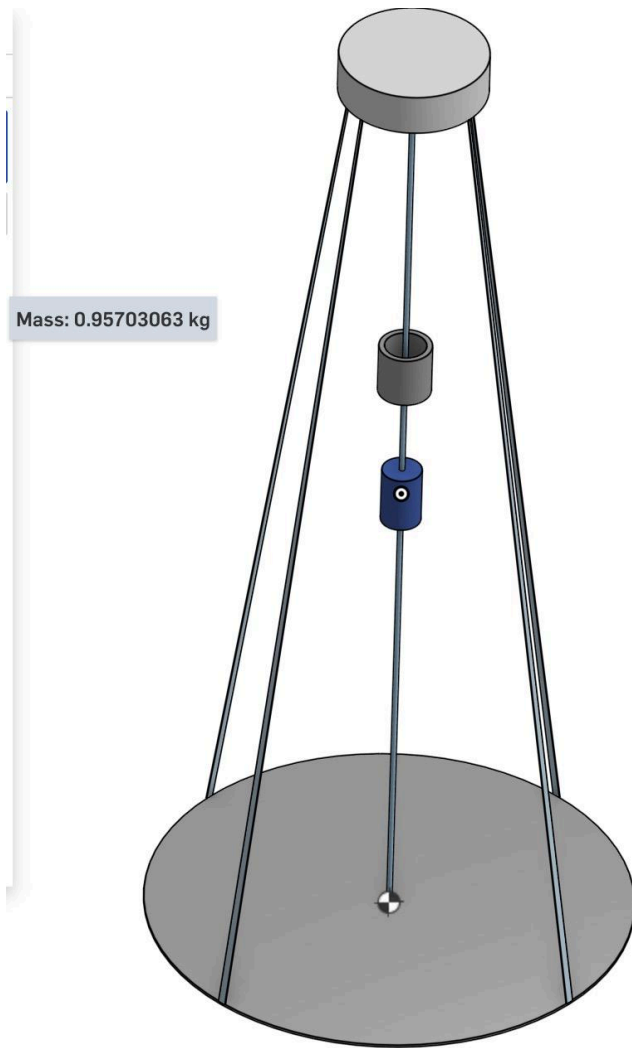


Figure F1: CAD mass tool screenshot, demonstrating structural approximations and total weight.

Appendix G) “Firestarter” Design Exploration

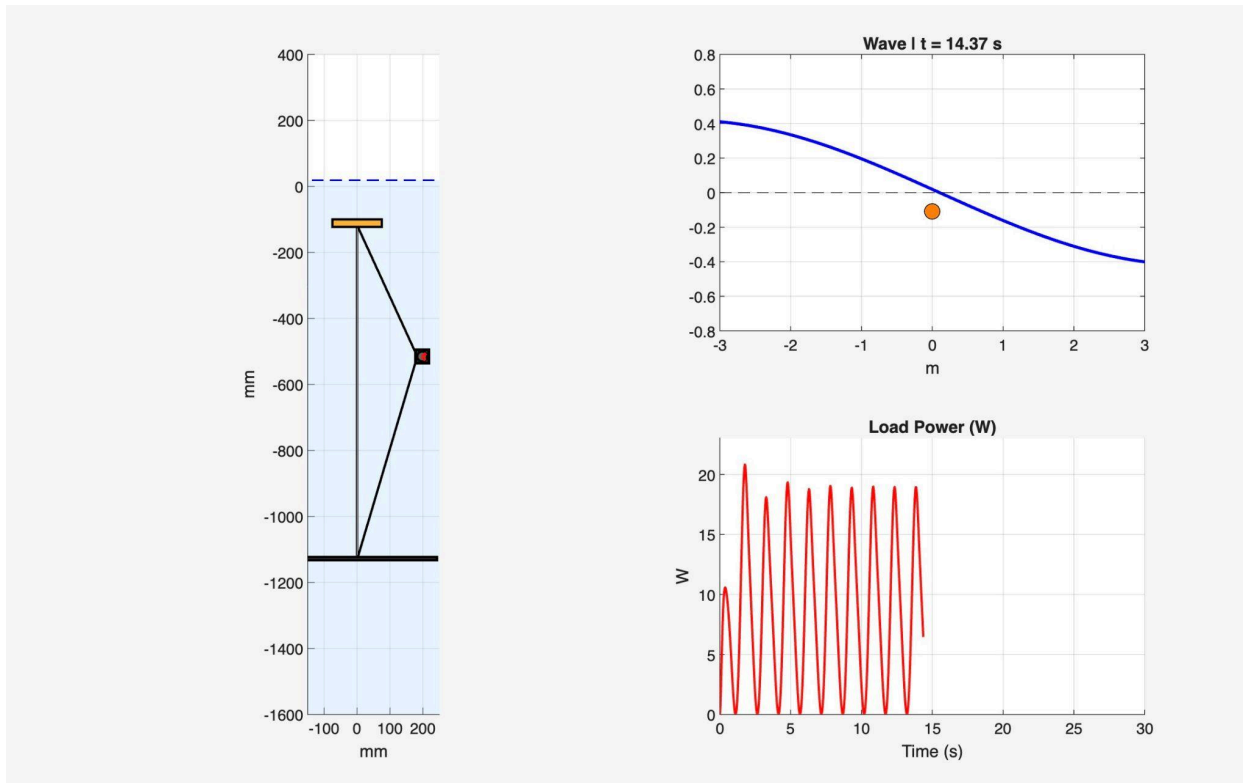


Figure G1: The calculations for a rotary generator driven by cords, and the resulting MATLAB output figures.

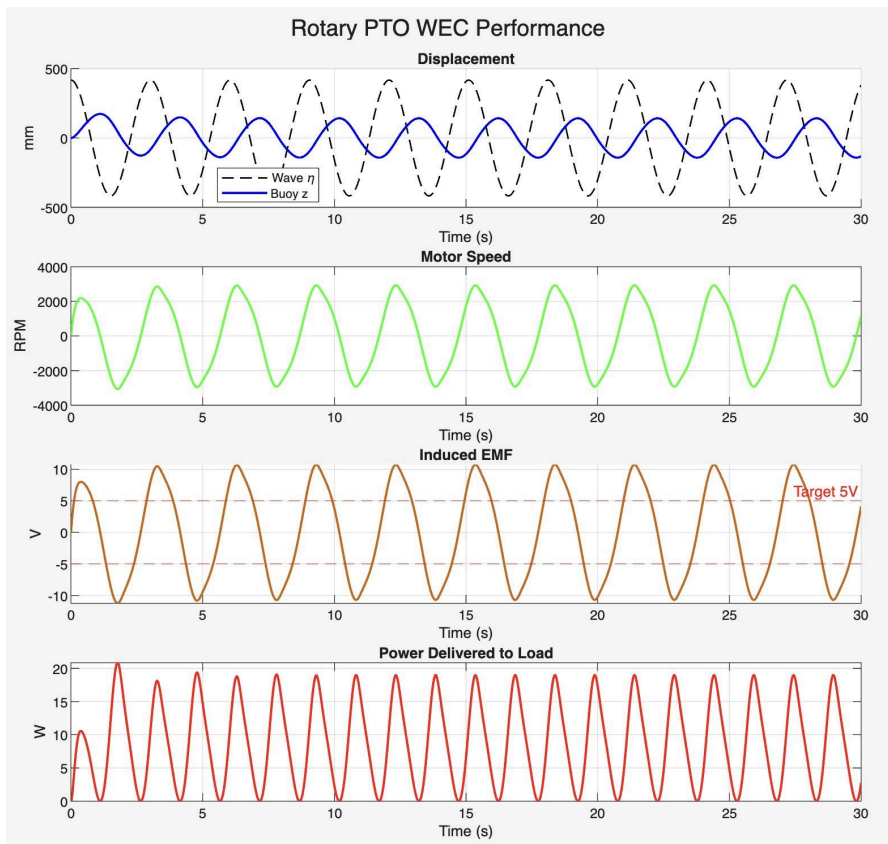


Figure G2: Output figures. It should be noted that the large energy density increase is over a direct drive linear generator